

Inversion

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When you were little, by looking at mirrors or otherwise, you learned about *reflections across a line*. In school, you also learned about *reflections across a point*. Today you'll learn what it means to *reflect about a circle*. This is known as *inversion*, and this operation (and its many variants) are very useful in olympiad geometry.

Much of today's lecture is similar to Chapter 8 from Evan Chen's Euclidean Geometry in Math Olympiads. Please read EGMO Chapter 8 (and EGMO in general) if you have not yet! There are some topics there that we will not discuss today, so it will still be a useful reference to you.

1 Inversion Review

Throughout, given a map f and a geometric object Γ , we say that f *preserves* Γ if $f(\Gamma) = \Gamma$. Notably f preserving Γ will not necessarily mean that f fixes every point on Γ .

Definition 1.1. Given a circle Ω with center O and radius r , *inversion* about Ω is the operation which takes a point P to the point Q on ray $\overrightarrow{OP}$ satisfying $OQ \times OP = r^2$.

This technically does not define where O goes to. But we can view it as going to the “point at infinity”, and we will make this rigorous soon.

Theorem 1.2. Consider a circle Ω with center O , and let ϕ be the operation given by inverting about Ω . Then ϕ has the following properties:

1. ϕ is an involution, i.e. $\phi \circ \phi = \text{id}$.
2. ϕ preserves lines passing through O .
3. ϕ maps lines not passing through O to circles passing through O , and vice versa.
4. ϕ maps circles not passing through O to other circles not passing through O .

Proof. 1) and 2) are clear. We'll do 3) and 4) in class. □

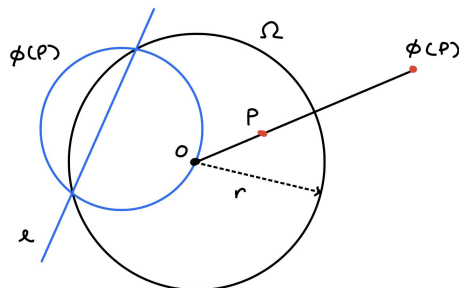


Figure 1: Here ϕ denotes inversion about the circle Ω .

Define a *cline* to be either a circle or a line. Then we can think of Theorem 1.2 as stating that inversions map clines to clines.

Inversion also takes the following nice form in complex coordinates.

Example 1.3. Let Ω be the unit circle in the complex plane. Then inversion about Ω maps

$$re^{i\theta} \mapsto e^{i\theta}/r = \overline{e^{-i\theta}/r} = \overline{1/(re^{i\theta})}.$$

Thus its formula is given by $z \mapsto 1/\bar{z}$.

In this sense we may view inversion as an involution on the *Riemann sphere* $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. This also formalizes the notion that 0 goes to the *complex point at infinity* $\infty \in \hat{\mathbb{C}}$.

2 Problem Solving with Inversion

Line and point reflections are useful concepts, but they cannot be too powerful for olympiad problems, since symmetries about a line / point are usually easy to spot. However, it is much harder to eyeball what a diagram looks like after inversion! This is what makes inversion so powerful: many problems which seem complicated become much easier after performing an inversion, so it gives a very useful alternative viewpoint to a problem. Notably, if a problem contains *a lot of circles passing through some point*, then inverting about that point will turn those circles into lines. This can be very useful, as we'll see in the following examples.

First, let's practice with some common configurations.

Example 2.1. Let ABC be an acute triangle, let H be its orthocenter, and let D , E , and F be the feet of the A , B , and C altitudes (respectively). Consider inversion about the circle centered at A with radius $\sqrt{AH \cdot AD}$. Where does this send $\{A, B, C, D, E, F, H\}$?

Example 2.2. Let ABC be a triangle, let I be its incenter, and let D , E , and F be the A , B , and C intouch points (respectively). Consider inversion about the incircle. Where does this send $\{A, B, C\}$? Where does this send (ABC) ?

Do one of the following two examples on your own:

Example 2.3. Let ABC be a triangle, let I be its incenter, and let I_A be its A -excenter. Consider inversion about the circle centered at A with radius $\sqrt{AB \cdot AC}$ (this is related to $\sqrt{bc}$ inversion, which we will learn later). Where does this send $\{B, C, I, I_A\}$?

Example 2.4. Let ABC be a triangle and let O be its circumcenter. Perform an inversion about a circle centered at A of any radius, and let asterisks denote images after inversion. How is O^* related to A , B^* , and C^* ?

Let's now practice applying inversion to contest problems. As previously discussed, it will be useful to invert about points with a lot of circles passing through them.

Example 2.5. Let $ABCD$ be a convex quadrilateral such that diagonals AC and BD intersect at right angles, and let E be their intersection. Prove that the reflections of E across AB , BC , CD , DA are concyclic.

1. Let ω_A be the circle centered at A passing through E , and define ω_B , ω_C , ω_D similarly. What points does ω_A pass through? What about the other three circles?
2. What happens when you invert about E ?

Here's a problem that would be very hard without inversion, but becomes much easier using it.

Example 2.6. [Ukraine NMO 2025 10.7] Given a triangle ABC , an arbitrary point D is chosen on the side AC . In triangles ABD and CBD , the angle bisectors BK and BL are drawn, respectively. The point O is the circumcenter of $\triangle KBL$. Prove that the second intersection point of the circumcircles of triangles ABL and CBK lies on the line OD .

1. Perform an inversion about the point B . (You can make this a $\sqrt{ac}$ inversion if you wish).
2. Let asterisks denote images after the above inversion. How is O^* related to B , K^* , and L^* ?
3. Finish the problem.

3 Orthogonality

When is a circle ω fixed under inversion about another circle Ω ?

Definition 3.1. Two circles Ω_1 and Ω_2 with centers O_1, O_2 (respectively) are *orthogonal* if the intersections P, Q of Ω_1 and Ω_2 satisfy $\angle O_1PO_2 = \angle O_1QO_2 = 90^\circ$.

Example 3.2. Prove that inversion about Ω fixes ω iff Ω and ω are orthogonal.

Orthogonality is a particularly useful condition because it is *reflexive*. Notably given two circles Ω, ω it implies that the inversion about Ω fixes ω iff the inversion about ω fixes Ω ! It also shows up in a lot of contexts, such as the following.

Example 3.3. Consider a circle Ω and a point P outside of Ω . Let the tangents from P to Ω touch Ω at R and S . Given any other point Q in the plane, prove that the circle with diameter PQ is orthogonal to Ω iff $Q \in RS$.

For example, this solves the Warmup G3 from earlier:

Example 3.4. Consider three circles $\Gamma_1, \Gamma_2, \Gamma_3$ which are non-overlapping and mutually external. For each point P in the plane outside the three circles, construct six points $A_1, B_1, A_2, B_2, A_3, B_3$ as follows: for each $i = 1, 2, 3$, A_i, B_i are distinct points on the circle Γ_i such that the lines PA_i and PB_i are both tangents to Γ_i . Call a point P exceptional if, from the construction, the three lines A_1B_1, A_2B_2, A_3B_3 are concurrent. Show that every exceptional point of the plane lies on a same circle.

4 Variants of Inversion

Here are two variants of inversion that show up quite often.

4.1 $\sqrt{bc}$ Inversion

Inversion is a great way for getting an alternate perspective on a problem. But what if we could look at both perspectives at once? This leads us to *$\sqrt{bc}$ inversion*, which is defined as follows.

Definition 4.1. Given a triangle ABC , $\sqrt{bc}$ inversion is the operation which inverts about the circle centered at A with radius $\sqrt{AB \cdot AC}$, then *reflects* across the angle bisector of $\angle BAC$.

Use asterisks to denote images after $\sqrt{bc}$ inversion. Then the point of $\sqrt{bc}$ inversion is that it maps $C^* = B$, $B^* = C$, so the three points $\{A, B, C\}$ are permuted by $\sqrt{bc}$ inversion. In this way, we may piece our original and $\sqrt{bc}$ -inverted diagrams together! This is known as an *overlay*.

Example 4.2. Determine where the following points / geometric objects go under $\sqrt{bc}$ inversion:

1. The incenter I , and the three excenters I_A, I_B, I_C .
2. The circumcenter O and the orthocenter H .
3. The circles (ABC) and (BIC) .
4. The midpoint M , and the two arc midpoints of arc BC .
5. The perpendicular bisector of BC .

For example, here's a problem which is demystified under $\sqrt{bc}$ inversion.

Example 4.3. Prove that there exists a point K on the circle with diameter AH , such that the circles (ABK) and (ACK) are tangent to BC . (K is known as the *A-HM point*).

4.2 Negative Inversion

One property of inversion is that given any circle Ω and point P lying *outside* Ω , there exists an inversion centered at P which fixes Ω . Indeed, take a tangent from P to Ω touching Ω at Q ; then inversion about (P, PQ) fixes Ω . But what if P lies inside Ω ? In this case, we may use a *negative inversion*, defined as follows:

Definition 4.4. Given a point P and a real $r > 0$, *negative inversion about P with radius r* maps any point Q to the point R such that $PQ \cdot PR = r^2$ and P lies on segment QR . Equivalently, it inverts about the circle centered at P with radius r , then reflects.

Here is a common scenario in which one encounters negative inversions.

Example 4.5. Consider an acute triangle ABC with orthocenter H . Let the feet of the A , B , and C altitudes respectively be D , E , and F .

1. Consider the negative inversion centered at H with radius $\sqrt{AH \cdot HD}$. Where does it send $\{B, C, E, F\}$? Where does it send (ABC) ?
2. Let $P \neq H$ be any point in the plane. Using this negative inversion, prove that the circles (ADP) , (BEP) , and (CFP) concur at a point other than P .

5 A Legendary Example

Here's a legendary example of an IMO problem that featured orthogonality. It doesn't quite require inversion as directly as the other problems, but it's a problem that requires you to understand inversion, and related techniques, very well.

Example 5.1. [IMO 2012 5] Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK . Show that $MK = ML$.

1. The $BK = BC$ condition is staring at us in the face. Construct K and L by drawing $\omega_A = (A, AC)$ and $\omega_B = (B, BK)$. Can you say anything about how X relates to ω_A and ω_B ? (See the next step if this doesn't make sense).
2. Let AK intersect ω_B again at K' , and let BK intersect ω_A again at L' . Prove that $KLK'L'$ is cyclic.
3. Draw $(KLK'L')$. Do you notice any tangencies it satisfies? Can you prove that these tangencies would finish the problem?
4. Prove that these tangencies are equivalent to showing certain pairs of circles are orthogonal, and show these orthogonalities.

6 Problems

6.1 Theroetical Warm-Ups

Here are some warmups to get you started.

1. Let a *quasi-harmonic* quadrilateral be a quadrilateral $ABCD$ satisfying $AB \cdot CD = AD \cdot BC$. What happens if you invert a quasi-harmonic quadrilateral about A ?
2. Consider two circles Ω_1, Ω_2 intersecting at points P and Q , and suppose their external tangents intersect at a point X . Prove that inversion about the circle (X, XP) swaps Ω_1 and Ω_2 .
3. Consider two circles Ω and ω , and suppose Ω has center O . A line through O intersects ω at P and Q . Prove that ω and Ω are orthogonal iff inversion about Ω swaps P and Q .
4. Given two points A, B and a circle Ω of radius r centered at O , use asterisks to denote images under inversion about Ω . Give a formula for the length of A^*B^* in terms of O, A, B , and r .
5. It turns out that $\sqrt{bc}$ inversions produces a lot of similarities.
 - (a) Express $\sqrt{bc}$ inversion in terms of complex numbers.
 - (b) Given any point $D \neq A$ in the plane, let E be its image under $\sqrt{bc}$ inversion. Prove $ADB \sim ACE$.
6. Consider an acute triangle ABC . Define its *A-mixtilinear incircle* Ω to be the circle tangent to AB and AC , and internally tangent to (ABC) .
 - (a) Where does the mixtilinear incircle go under $\sqrt{bc}$ -inversion?
 - (b) The *A-mixtilinear touchpoint* T is the tangency point of Ω and (ABC) . Let E be the A -extouch point. Prove that $\angle BAT = \angle CAE$.

6.2 Contest Practice

6.3 Part A

Here are some IMO 1 level problems.

1. Two circles intersect at two points A and B . A line l passing through A meets these circles again at C and D , such that C, D lie on the arcs AB of the two circles which are not contained in the other circle. Let M, N respectively be the midpoints of the arcs BC, BD (not containing A) of their respective circles. Let K be the midpoint of segment CD . Prove $\angle MKN = 90^\circ$.
2. Let Ω be the circumcircle of the triangle ABC . The circle ω is tangent to the sides AC and BC , and it is internally tangent to the circle Ω at the point P . A line parallel to AB intersecting the interior of triangle ABC is tangent to ω at Q .
Prove that $\angle ACP = \angle QCB$.
3. Let Γ be the circumcircle of triangle ABC . A circle Ω is tangent to the line segment AB and is tangent to Γ at a point lying on the same side of the line AB as C . The angle bisector of $\angle BCA$ intersects Ω at two different points P and Q . Prove that $\angle ABP = \angle QBC$.
4. ★ Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P , and Γ_2, Γ_4 are externally tangent at the same point P . Suppose that Γ_1 and Γ_2 ; Γ_2 and Γ_3 ; Γ_3 and Γ_4 ; Γ_4 and Γ_1 meet at A, B, C, D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

5. Let AB be the diameter of a circle Γ and let C be a point on Γ different from A and B . Let D be the foot of perpendicular from C on to AB . Let K be a point on the segment CD such that AC is equal to the semi perimeter of ADK . Show that the A -excircle of ADK is tangent to Γ .
6. ★ Consider a regular hexagon $A_1A_2A_3A_4A_5A_6$, and a point P in the plane which is not on any of A_1A_4, A_2A_5, A_3A_6 . Prove that the circles $(PA_1A_4), (PA_2A_5), (PA_3A_6)$ concur at a point distinct from P .
7. Let ABC be a triangle with circumcircle Ω and A -excenter J . The incircle of triangle ABC touches its side BC at D . Let K, L be the two intersection points of Ω and the circle with diameter DJ , and lines DK and DL meet Ω again at P and Q , respectively. Prove that line PQ passes through the incenter of triangle ABC .
8. Let ω_1 and ω_2 be two orthogonal circles, and let the center of ω_1 be O . Diameter AB of ω_1 is selected so that B lies strictly inside ω_2 . The two circles tangent to ω_2 , passing through O and A , touch ω_2 at F and G . Prove that $FGOB$ is cyclic.
9. In a convex quadrilateral $ABCD$, $\angle ABC = \angle CDA$. Let $X \neq C$ be the intersection of the circumcircle of $\triangle BCD$ and circle with diameter AC . Prove that the tangent to the circumcircle of $\triangle BCD$ at X , the tangent to the circumcircle of $\triangle ABD$ at A concur on BD .
10. Let A, B, C, D be four points in the plane, with C and D on the same side of the line AB , such that $AC \cdot BD = AD \cdot BC$ and $\angle ADB = 90^\circ + \angle ACB$. Find the ratio

$$\frac{AB \cdot CD}{AC \cdot BD},$$

and prove that the circumcircles of the triangles ACD and BCD are orthogonal.

6.4 Part B

Here are some IMO 1-2 level problems.

1. Let $ABCD$ be a cyclic quadrilateral with circumcenter O . Let P be the intersection of AC and BD . Points X and Y are on the plane such that $PX \parallel AB$ and $PY \parallel CD$. It is also known that AC bisects $\angle XCD$ and BD bisects $\angle ABY$. Prove that $XY \perp OP$.
2. Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.
3. ★ Consider three mutually intersecting circles $\Omega_1, \Omega_2, \Omega_3$. Suppose Ω_i and Ω_{i+1} share a common chord l_{i+2} , and these three common chords intersect at a point P . Also suppose there exists two circles ω, Ω internally tangent to all three circles.
 - (a) Prove that P is either the exsimilicenter or insimilicenter of ω and Ω .
 - (b) Which center is it?
4. Let ABC be an acute scalene triangle with circumcenter O , and let T be on line BC such that $\angle TAO = 90^\circ$. The circle with diameter $\overline{AT}$ intersects the circumcircle of $\triangle BOC$ at two points A_1 and A_2 , where $OA_1 < OA_2$. Points B_1, B_2, C_1, C_2 are defined analogously. Prove that $\overline{AA_1}, \overline{BB_1}, \overline{CC_1}$ are concurrent. Prove that $\overline{AA_2}, \overline{BB_2}, \overline{CC_2}$ are concurrent on the Euler line of triangle ABC .
5. ★ Five points A, B, C, D and E lie on a circle τ clockwise in that order such that $AB \parallel CE$ and $\angle ABC > 90^\circ$. Let k be a circle tangent to AD, CE and τ such that k and τ touch on the arc $\widehat{DE}$ not containing A, B and C . Let $F \neq A$ be the intersection of τ and the tangent line to k passing through A different from AD .
Prove that there exists a circle tangent to BD, BF, CE and τ .

6. Let $B = (-1, 0)$ and $C = (1, 0)$ be fixed points on the coordinate plane. A nonempty, bounded subset S of the plane is said to be nice if
- (i) there is a point T in S such that for every point Q in S , the segment TQ lies entirely in S ; and
 - (ii) for any triangle $P_1P_2P_3$, there exists a unique point A in S and a permutation σ of the indices $\{1, 2, 3\}$ for which triangles ABC and $P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$ are similar.
- Prove that there exist two distinct nice subsets S and S' of the set $\{(x, y) : x \geq 0, y \geq 0\}$ such that if $A \in S$ and $A' \in S'$ are the unique choices of points in (ii), then the product $BA \cdot BA'$ is a constant independent of the triangle $P_1P_2P_3$.
7. ★ Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.
8. ★ Triangle ABC has incircle ω and A -excircle ω_A . Circle γ_B passes through B and is externally tangent to ω and ω_A . Circle γ_C passes through C and is externally tangent to ω and ω_A . If γ_B intersects line BC again at D , and γ_C intersects line BC again at E , prove that $BD = EC$.
9. Let $\triangle ABC$ be an acute-angled triangle with altitudes AD and BE meeting at H . Let M be the midpoint of segment AB , and suppose that the circumcircles of $\triangle DEM$ and $\triangle ABH$ meet at points P and Q with P on the same side of CH as A . Prove that the lines ED, PH , and MQ all pass through a single point on the circumcircle of $\triangle ABC$.
10. Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different and lie on Γ in this order.
- Prove that the circumcircles of triangles KQH and FKM are tangent to each other.

6.5 Part C

Here are four IMO 3 level problems. They're quite challenging, so good luck!

1. Elmo is now learning olympiad geometry. In triangle ABC with $AB \neq AC$, let its incircle be tangent to sides BC, CA , and AB at D, E , and F , respectively. The internal angle bisector of $\angle BAC$ intersects lines DE and DF at X and Y , respectively. Let S and T be distinct points on side BC such that $\angle XSY = \angle XTY = 90^\circ$. Finally, let γ be the circumcircle of $\triangle AST$.
 - (a) Help Elmo show that γ is tangent to the circumcircle of $\triangle ABC$.
 - (b) Help Elmo show that γ is tangent to the incircle of $\triangle ABC$.
2. Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangle PCE and PBF meet again at Q . Prove that lines DI and PQ meet on the line through A perpendicular to AI .
3. Let I be the incentre of a non-equilateral triangle ABC , I_A be the A -excentre, I'_A be the reflection of I_A in BC , and l_A be the reflection of line AI'_A in AI . Define points I_B, I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B . Prove that P lies on line OI where O is the circumcentre of triangle ABC . Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y . Show that $\angle XIY = 120^\circ$.
4. ★ A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.